

Geofencing Location Event Detection

Admin
System administrator

Dashboard

Real-time geofence intelligence and device activity.



Recent device activity

The most recent geofence actions across your fleet.

Live

DATE & TIME	DEVICE	GEOFENCE	ACTION
10/2/2026, 3:37:46 AM	Air Device	Tuticorin City Zone	OUTSIDE
10/2/2026, 3:37:46 AM	Air Device	Airport Zone	INSIDE
10/2/2026, 3:37:46 AM	Air Device	Railway Zone	OUTSIDE
10/2/2026, 3:37:46 AM	Air Device	Harbour Zone	OUTSIDE
10/2/2026, 3:35:55 AM	Ship Device	Tuticorin City Zone	OUTSIDE
10/2/2026, 3:35:55 AM	Ship Device	Airport Zone	OUTSIDE
10/2/2026, 3:35:55 AM	Ship Device	Railway Zone	OUTSIDE
10/2/2026, 3:35:55 AM	Ship Device	Harbour Zone	ENTER

Geofencing Location Event Detection

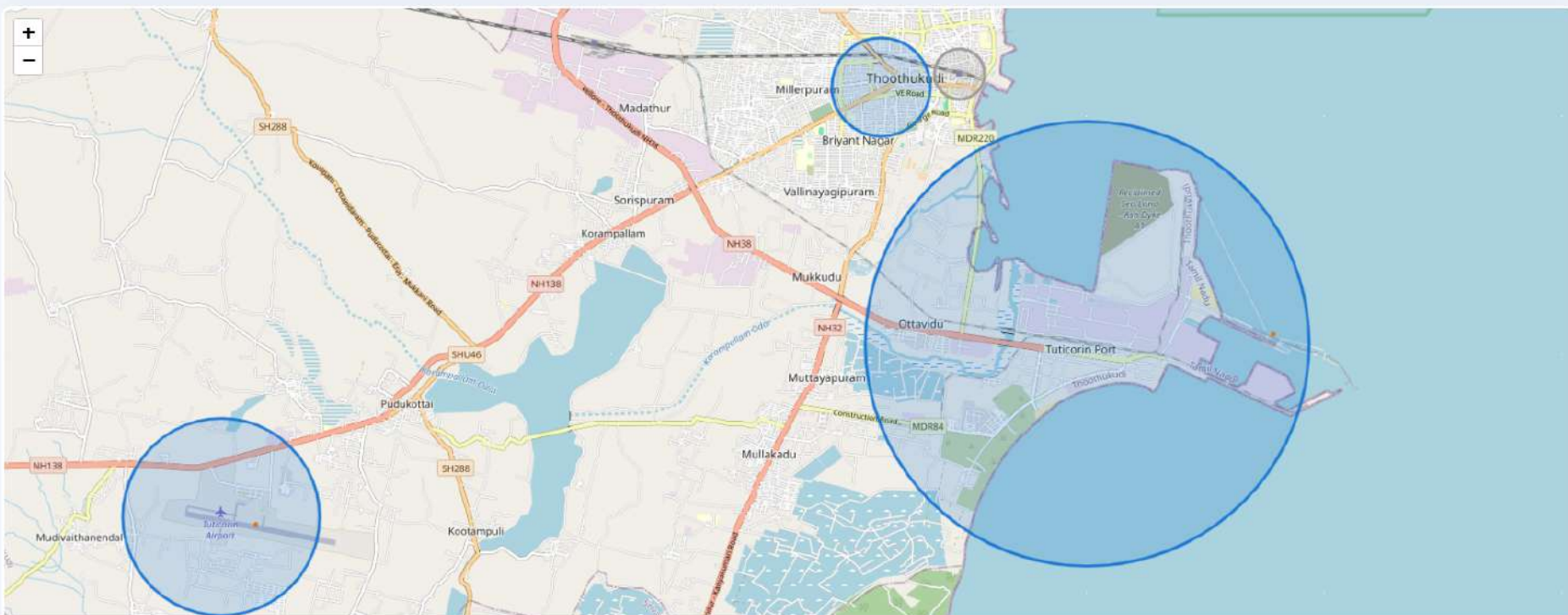
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Live map

Live geofence boundaries and tracked-device positions.

2 ACTIVE DEVICES

Map data refreshes every 15 seconds. Green markers indicate devices inside their latest evaluated geofence.



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Create geofence

Name*
Airport Zone

Boundary type
Circle

Center latitude*
8.725179

Center longitude*
78.018802

Radius (meters)*
3000

GPS accuracy buffer (meters)
15

Tracking update interval (seconds)*
60

Unchanged INSIDE or OUTSIDE statuses are recorded at this interval. ENTER and EXIT are always recorded immediately.

Event rules

- ☒ enter
- ☒ exit
- ☒ inside
- ☒ outside

Geofencing Location Event Detection

A

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Geofences

Manage protected zones, event rules, and boundary coverage.

4 GEOFENCES

+ Create geofence

4
Total geofences

3
Active boundaries

4
Circular zones

🔍 Search geofences

⚙ All statuses

GEOFENCE	BOUNDARY	EVENT RULES	TRACKING	STATUS	ACTIONS
Airport Zone Radius boundary	2000 m radius	EXIT ENTER INSIDE OUTSIDE	60s	ACTIVE	 
Harbour Zone Radius boundary	4500 m radius	EXIT ENTER INSIDE OUTSIDE	60s	ACTIVE	 
Railway Zone Radius boundary	500 m radius	EXIT ENTER INSIDE OUTSIDE	60s	INACTIVE	 
Tuticorin City Zone Radius boundary	1000 m radius	EXIT ENTER INSIDE OUTSIDE	60s	ACTIVE	 

GeoSystem

Dashboard

Live map

Geofences

Devices

Event history

Audit logs

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2

2

Registered devices

2

2

Located devices

2 DEVICES

+ Add device

Search device name or ID

All statuses

DEVICE	DEVICE ID	STATUS	ACTIONS
Air Device	D-25-001	ACTIVE	
Ship Device	D-26-002	ACTIVE	

Add device

Device name*

Road Device

Device ID*

D-26-002

Latitude

8.801840

Longitude

78.138296

Cancel

Save device

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Devices

Register, monitor, and manage tracked devices.

3 DEVICES

+ Add device

3 Registered devices

3 3
Active devices

3 3
Located devices

🔍 Search device name or ID

≡ All statuses ▾

Device	Device ID	Location Status	Status	Actions
Air Device	D-25-001	View on Map	Active	Edit Delete
Road Device	D-26-002	View on Map	Active	Edit Delete
Ship Device	D-26-001	View on Map	Active	Edit Delete

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Live map

Live geofence boundaries and tracked-device positions.

3 ACTIVE DEVICES

Map data refreshes every 15 seconds. Green markers indicate devices inside their latest evaluated geofence.



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Event history

A complete, time-ordered record of geofence activity.

10 EVENTS RECORDED

DATE & TIME	DEVICE	GEOFENCE	EVENT	TRANSITION
10/2/2026, 4:03:04 AM	Road Device D-26-002	Tuticorin City Zone	INSIDE	UNKNOWN → INSIDE
10/2/2026, 4:03:04 AM	Road Device D-26-002	Airport Zone	OUTSIDE	UNKNOWN → OUTSIDE
10/2/2026, 4:03:04 AM	Road Device D-26-002	Harbour Zone	OUTSIDE	UNKNOWN → OUTSIDE
10/2/2026, 3:37:46 AM	Air Device D-25-001	Tuticorin City Zone	OUTSIDE	UNKNOWN → OUTSIDE
10/2/2026, 3:37:46 AM	Air Device D-25-001	Airport Zone	INSIDE	UNKNOWN → INSIDE
10/2/2026, 3:37:46 AM	Air Device D-25-001	Railway Zone	OUTSIDE	UNKNOWN → OUTSIDE
10/2/2026, 3:37:46 AM	Air Device D-25-001	Harbour Zone	OUTSIDE	UNKNOWN → OUTSIDE
10/2/2026, 3:35:55 AM	Ship Device D-26-001	Tuticorin City Zone	OUTSIDE	outside → OUTSIDE
10/2/2026, 3:35:55 AM	Ship Device D-26-001	Airport Zone	OUTSIDE	outside → OUTSIDE

10/2/2026, 3:37:46 AM	Air Device D-25-001	Railway Zone	OUTSIDE	UNKNOWN – OUTSIDE
10/2/2026, 3:37:46 AM	Air Device D-25-001	Harbour Zone	OUTSIDE	UNKNOWN – OUTSIDE
10/2/2026, 3:35:55 AM	Ship Device D-26-001	Tuticorin City Zone	OUTSIDE	outside → OUTSIDE
10/2/2026, 3:35:55 AM	Ship Device D-26-001	Airport Zone	OUTSIDE	outside → OUTSIDE
10/2/2026, 3:35:55 AM	Ship Device D-26-001	Railway Zone	OUTSIDE	outside → OUTSIDE
10/2/2026, 3:35:55 AM	Ship Device D-26-001	Harbour Zone	ENTER	outside → INSIDE
10/2/2026, 3:34:27 AM	Ship Device D-26-001	Tuticorin City Zone	EXIT	inside → OUTSIDE
10/2/2026, 3:34:27 AM	Ship Device D-26-001	Airport Zone	OUTSIDE	UNKNOWN – OUTSIDE
10/2/2026, 3:34:27 AM	Ship Device D-26-001	Railway Zone	OUTSIDE	UNKNOWN – OUTSIDE
10/2/2026, 3:34:27 AM	Ship Device D-26-001	Harbour Zone	OUTSIDE	UNKNOWN – OUTSIDE
10/1/2026, 1:32:18 PM	Ship Device D-26-001	Tuticorin City Zone	INSIDE	inside – INSIDE
10/1/2026, 1:31:01 PM	Ship Device D-26-001	Tuticorin City Zone	INSIDE	inside – INSIDE
10/1/2026, 12:57:11 PM	Ship Device D-26-001	Tuticorin City Zone	INSIDE	UNKNOWN – INSIDE

Geofencing API 1.0.0 OAS 3.1

/openapi.json

geofences



GET	/api/geofences	List Geofences	 
POST	/api/geofences	Create Geofence	
GET	/api/geofences/{fence_id}	Read Geofence	
PUT	/api/geofences/{fence_id}	Update Geofence	
PATCH	/api/geofences/{fence_id}	Patch Geofence	
DELETE	/api/geofences/{fence_id}	Delete Geofence	
PATCH	/api/geofences/{fence_id}/status	Set Geofence Status	

devices



GET	/api/devices	List Devices	
POST	/api/devices	Create Device	
PUT	/api/devices/{device_id}	Update Device	

DELETE /api/geofences/{fence_id} Delete Geofence

PATCH /api/geofences/{fence_id}/status Set Geofence Status

devices

GET /api/devices List Devices

POST /api/devices Create Device

PUT /api/devices/{device_id} Update Device

GET /api/devices/{device_id} Read Device

PATCH /api/devices/{device_id} Patch Device

DELETE /api/devices/{device_id} Delete Device

PATCH /api/devices/{device_id}/status Set Device Status

GET /api/devices/{device_id}/events Device Events

location tracking

GET /api/locations/{device_id}/latest Latest Location

GET /api/locations/{device_id}/history Location History

POST /api/locations Ingest Location

GET /api/devices/{device_id}/events Device Events

location tracking

GET /api/locations/{device_id}/latest Latest Location

GET /api/locations/{device_id}/history Location History

POST /api/locations Ingest Location

event history

GET /api/events List Events

GET /api/events/summary Event Summary

GET /api/events/{event_id} Read Event

audit logs

GET /api/audit-logs List Audit Logs

default

GET /health Health

Schemas

GET /api/geofences List Geofences

Parameters Cancel

No parameters

Execute

Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/api/geofences' \
  -H 'accept: */*'

```

Request URL

```
http://127.0.0.1:8000/api/geofences

```

Server response

Code	Details
200	Response body <pre>[{ "name": "Airport Zone", "boundary_type": "circle", "enabled": true, "center_latitude": 8.72518, "center_longitude": 78.0188, "radius_meters": 2000, "accuracy_buffer_meters": 15, "tracking_interval_seconds": 60, "event_rules": { "exit": true, "enter": true, "inside": true, "outside": true }, "points": [], "id": 2 }]</pre>

devices

GET /api/devices List Devices

Parameters

No parameters

Execute Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/api/devices' \
  -H 'accept: */*'
```

Request URL

```
http://127.0.0.1:8000/api/devices
```

Server response

Code	Details
200	Response body <pre>[{ "external_id": "0-25-001", "label": "Air Device", "user_id": null, "enabled": true, "latitude": 8.72382, "longitude": 78.0252, "id": 2 }, { "external_id": "0-26-002",</pre>

GET /api/locations/{device_id}/latest Latest Location

Parameters

Cancel

Name	Description
device_id * required integer (path)	1

Execute

Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/api/locations/1/latest' \
  -H 'accept: */*'

```

Request URL

```
http://127.0.0.1:8000/api/locations/1/latest

```

Server response

Code	Details
200	Response body <pre>{ "id": 6, "device_id": 1, "latitude": 8.7585, "longitude": 78.2131, "recorded_at": "2026-10-02T03:36:43", "received_at": "2026-10-02T09:06:42" } </pre> Download

Response headers

GET /api/events/summary Event Summary

Parameters Cancel

No paramotors

Execute

Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/api/events/summary' \
  -H 'accept: */*'
```

Request URL

```
http://127.0.0.1:8000/api/events/summary
```

Server response

Code	Details
200	Response body <pre>{ "total": 19, "enter": 1, "exit": 1, "inside": 5, "outside": 11 }</pre> Download Response headers <pre>content-length: 55 content-type: application/json date: Fri, 02 Oct 2025 04:09:33 GMT server: uvicorn</pre>

Name Description

fence_id * required

integer
(path)

1

Execute

Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/api/geofences/1' \
  -H 'accept: */*'
```

Request URL

http://127.0.0.1:8000/api/geofences/1

Server response

Code Details

200

Response body

```
{
  "name": "Tuticorin City Zone",
  "boundary_type": "circle",
  "enabled": true,
  "center_latitude": 8.80359,
  "center_longitude": 78.1408,
  "radius_meters": 1000,
  "accuracy_buffer_meters": 15,
  "tracking_interval_seconds": 60,
  "event_rules": {
    "exit": true,
    "enter": true,
    "inside": true,
    "outside": true
  },
  "points": [],
  "id": 1
}
```

Download

GET /api/devices/{device_id} Read Device

Parameters

Cancel

Name	Description
device_id * required integer (path)	3

Execute

Clear

Responses

Curl

```
curl -X 'GET' \  
  'http://127.0.0.1:8000/api/devices/3' \  
  -H 'accept: */*'
```

Request URL

```
http://127.0.0.1:8000/api/devices/3
```

Server response

Code	Details
200	Response body <pre>{ "external_id": "D-26-002", "label": "Road Device", "user_id": null, "enabled": true, "latitude": 8.80184, "longitude": 78.1383, "id": 3</pre> Download